Solving Quadratics: Factoring, Graphs, and the Zero Product Property

Intermediate Algebra Worksheet · Grade 10-12 · numberbender.com

Name: ______________________ Date: ______________ Score: / 10

Learning Objectives

- Identify quadratic equations in standard form and review factoring techniques
- Analyze the graph of a parabola including vertex, axis of symmetry, and intercepts
- Solve quadratic equations by factoring and applying the zero product property
- Determine the roots/solutions of quadratic equations and interpret them graphically

Solve each problem completely, showing all work where appropriate, and write your final answer in simplest form.

1. Factor the following trinomial completely (review of factoring).

$$x^2 + 7x + 12$$

Answer: ______________________

2. Determine whether the following equation is a quadratic equation. Justify your answer.

$$3x^2 - 5x + 2 = 0$$

Answer: ______________________

3. Identify the values of a, b, and c for the quadratic equation written in standard form.

$$-2x^2 + 4x - 9 = 0$$

Answer: ______________________

4. Describe the graph of the quadratic function: state whether the parabola opens upward or downward and give its vertex.

$$f(x) = (x - 2)^2 + 3$$

Answer: ______________________

5. Solve the quadratic equation by factoring.

$$x^2 - 9x + 20 = 0$$

Answer: ______________________

6. Apply the zero product property to find the solutions of the equation.

$$(2x - 3)(x + 5) = 0$$

Answer: ______________________

7. Find the roots (solutions) of the quadratic equation by factoring.

$$x^2 - 16 = 0$$

Answer: ______________________

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8. Analyze the graph: find the x-intercepts (roots) of the quadratic function by factoring.

$f(x) = x^2 + 2x - 15$

Answer: ____________________________________

9. Solve the quadratic equation by first factoring out the greatest common factor, then applying the zero product property.

$2x^2 - 8x = 0$

Answer: ____________________________________

10. Find the axis of symmetry and the vertex of the parabola (as you would when analyzing a quadratic in Desmos).

$f(x) = x^2 - 6x + 5$

Answer: ____________________________________

This worksheet covers all topics from the lesson: review of factoring, introduction to and identification of quadratic equations, graphs of parabolas, solutions/roots of quadratics, analyzing quadratic graphs (vertex, axis of symmetry, intercepts), solving quadratics by factoring, the zero product property, and using technology (Desmos) to find roots.

Solutions

1. Factor the following trinomial completely (review of factoring).

$x^2 + 7x + 12$

→ *Find two numbers that multiply to twelve and add to seven.*
→ *The numbers three and four satisfy both conditions.*
→ *Write the factored form as the product of two binomials.*

Answer: $(x + 3)(x + 4)$

2. Determine whether the following equation is a quadratic equation. Justify your answer.

$3x^2 - 5x + 2 = 0$

→ *A quadratic equation has the standard form a x squared plus b x plus c equals zero, where a is not zero.*
→ *Here the highest power of x is two and the coefficient of x squared is three, which is not zero.*
→ *Therefore the equation is quadratic with a equals three, b equals negative five, and c equals two.*

Answer: Yes, it is quadratic with $a = 3,\ b = -5,\ c = 2$

3. Identify the values of a, b, and c for the quadratic equation written in standard form.

$-2x^2 + 4x - 9 = 0$

→ *Compare the equation to the standard form a x squared plus b x plus c equals zero.*
→ *The coefficient of x squared is negative two, so a equals negative two.*
→ *The coefficient of x is four, so b equals four, and the constant term is negative nine, so c equals negative nine.*

Answer: $a = -2,\ b = 4,\ c = -9$

4. Describe the graph of the quadratic function: state whether the parabola opens upward or downward and give its vertex.

$f(x) = (x - 2)^2 + 3$

→ *The function is written in vertex form, where the vertex is at the point h comma k.*
→ *Here h equals two and k equals three, so the vertex is at the point two comma three.*
→ *Because the leading coefficient is positive one, the parabola opens upward.*

Answer: Opens upward; vertex $(2, 3)$

5. Solve the quadratic equation by factoring.

$x^2 - 9x + 20 = 0$

→ *Find two numbers that multiply to twenty and add to negative nine.*
→ *The numbers negative four and negative five work, so the factored form is the quantity x minus four times the quantity x minus five equals zero.*
→ *By the zero product property, set each factor equal to zero to get x equals four or x equals five.*

Answer: $x = 4,\ x = 5$

6. Apply the zero product property to find the solutions of the equation.

$(2x-3)(x+5)=0$

→ *The zero product property states that if a product equals zero, then at least one factor must equal zero.*
→ *Set two x minus three equal to zero and solve to get x equals three halves.*
→ *Set x plus five equal to zero and solve to get x equals negative five.*

Answer: $x=\frac{3}{2},\ x=-5$

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7. Find the roots (solutions) of the quadratic equation by factoring.

$x^2-16=0$

→ *Recognize the expression as a difference of two squares.*
→ *Factor it as the quantity x minus four times the quantity x plus four equals zero.*
→ *By the zero product property, the solutions are x equals four and x equals negative four.*

Answer: $x=\pm 4$

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8. Analyze the graph: find the x-intercepts (roots) of the quadratic function by factoring.

$f(x)=x^2+2x-15$

→ *Set the function equal to zero to find the x-intercepts.*
→ *Factor the trinomial into the quantity x plus five times the quantity x minus three.*
→ *Apply the zero product property to get x equals negative five and x equals three as the x-intercepts.*

Answer: $x=-5,\ x=3$

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9. Solve the quadratic equation by first factoring out the greatest common factor, then applying the zero product property.

$2x^2-8x=0$

→ *Factor out the greatest common factor of two x from both terms to get two x times the quantity x minus four equals zero.*
→ *Set each factor equal to zero using the zero product property.*
→ *Solving gives x equals zero or x equals four.*

Answer: $x=0,\ x=4$

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10. Find the axis of symmetry and the vertex of the parabola (as you would when analyzing a quadratic in Desmos).

$f(x)=x^2-6x+5$

→ *The axis of symmetry is found using x equals negative b divided by two a.*
→ *Substitute a equals one and b equals negative six to get x equals three as the axis of symmetry.*
→ *Substitute x equals three into the function to find the y-coordinate of the vertex: nine minus eighteen plus five equals negative four, so the vertex is three comma negative four.*

Answer: Axis: $x=3$; Vertex: $(3,-4)$